ROUTE: Mapping a Mutational Ribozyme Fitness Landscape

RouteID: idk
Wall: Finals

Setter: Tania Frank
Grade: 5.10
Estimated time: 30-50 min
Prereqs: Python, being a cool cat

The Setup

In this route you will analyze variants of a ribozyme- an RNA enzyme. The ribozyme you will be analyzing self-cleaves through an inline attack of a 2' OH on an adjacent phosphate in the phosphodiester backbone, similar to hammerhead and hairpin ribozymes. This dataset comes from a high-throughput mutational scan of the Osa-1-4 twister ribozyme, where every single and double nucleotide mutant was synthesized, selected for self-cleaving activity, and read out by deep sequencing. Each variant has a relative activity number attached to it, and we want to understand what nucleotides are important for the structure and function of the ribozyme. It is a 54 nt sequence, and 48 were mutated.

Your job on this route is to turn that big table of mutants into a map of function along the ribozyme sequence. You will (1) quantify how sensitive each position is to mutation, (2) compute simple epistasis scores for double mutants, and (3) highlight pairs of nucleotides whose mutations correlate with each other– hints of base pairs or tertiary contacts. This is exactly the logic behind many current structure-inference and evolution-of-function papers.

Vocabulary

Ribozyme: RNA enzyme

Purine: A, G (2-ring nitrogenous bases)

Pyrimidine: C, U (1-ring nitrogenous bases)

Epistasis: Non-additive interaction between mutations

Relative Activity (RA): Activity of mutant ÷ wild-type activity

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Your Mission

1. Load the mutational matrix and reconstruct wild-type sequence
2. Map per-position mutational sensitivity from single mutants
3. Analyze purine vs. pyrimidine tolerance patterns
4. Compute epistasis scores for double mutants

5. Visualize covariant pairs as a position-pair matrix
6. Connect findings to known secondary structure

Data Files

Download the excel file of the mutational map: https://yokobayashilab.net/data.html

- Dataset: *2D ribozyme activity mapping of Osa-1-4 twister ribozyme* (Kobori & Yokobayashi, Angew. Chem. Int. Ed. 2016).
- You'll work from a provided Excel/CSV file containing the complete matrix of relative activities for all single and double mutants of a 48-nt core.

Exercise 0: Load the Data

Goal: Understand the matrix format and reconstruct wild-type sequence

1. Inspect the excel file, how is the data stored? Identify how variants are labeled (e.g., "A7U", "C15A") and where the wild-type activity is encoded
2. Load the excel file into your workspace

   Hint: modifying the excel file may make it easier to work with, such as a .csv

3. There are 6 nucleotides of the ribozyme at the 5' end that were not mutated: 5' CCGCCU 3' Using only the mutational matrix and this 5' region, what was the original RNA sequence? Store this.

   Hint: You could do it by hand but I discourage that. The data set only stores mutants. Remember, RNA and DNA are written in 5' to 3' directions.

Success Check:

- You understand what each value in the matrix represents
- You have the WT (wild type) RNA sequence
- You have a dataframe that contains every mutational variant and its relative activity

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Exercise 1: Single-mutant sensitivity–which positions matter?

Goal: map the conservation of individual nucleotides

Here you compress all single mutants down to a score for each position: how much does that position "tolerate" change?

Step 1: Extract single mutants

1. From the full table, extract the single-mutant entries (mutations at exactly one position relative to wild type)

Step 2: Define the mutational sensitivity at each position

1. We measure sensitivity at each position in 3 ways: for each position calculate the mean activity of the single mutants, the minimum relative activity of a single mutant, and a 'sensitivity score' defined as (1-mean relative activity)
2. Store these three values for each position in the RNA sequence

Step 3: Plot it

1. Plot these three values across the RNA sequence (position on x: axis, mutational sensitivity on y-axis and compare the graphs. Are they consistent?
2. Identify the 10 most sensitive nucleotides.

Questions: Are those three values consistent? What does each tell you about the position? In the literature there are 10 highly conserved NT: A7, A8, G23, G24, A28, A29, G30, C32, G45, A46. Does this align with your findings?

Success Check: you should have a ranked list of nucleotide sensitivity by position and plots describing which individual nucleotides are most conserved for ribozyme activity.

Exercise 2: Nucleotide identity and characteristics

Goal: Identify the importance of purines vs pyramidines and non-canonical base pairs at individual nucleotides positions.

We know that different nucleotides have different chemical structures and behaviors. Connecting those variations to activity can give information about the structural and functional importance of each nucleotide. Adenine and Guanine are purines, Uracil and Cytosine are pyrimidines. Besides the canonical watson crick base pairs (A-U, C-G), many ribozymes contain NC (non-canonical) GU pairs.

1. For each position compare the average Purine RA vs the average Pyrimidine RA.
2. Rank all C->U single mutants by RA.
   Hint: Guanosines readily form base pairs with Uridines…

Questions: Which positions tolerate any purine/pyrimidine equally? Which positions are purine-specific or pyrimidine-specific? Are there any C->U mutations that maintain strong activity? What may that hint about that nucleotide's secondary structure?

Success Check: A list of average Purine RA and average Pyrimidine activity by position, a ranked list of C->U mutations by RA

Exercise 3: Double mutants and epistasis

Goal: identify covariant nucleotide pairs

Ribozymes fold into active structures, which means that there are secondary and tertiary contacts across the RNA sequence. Structural methods like NMR and cryo-EM can reveal these architectures, but they are technically demanding and not always feasible for every RNA. Here, we instead analyze genetic interactions: by measuring how combinations of nucleotide substitutions affect activity, we use epistasis between mutations to pinpoint pairs and networks of residues whose interactions are critical for forming the active ribozyme structure and supporting its catalytic function.

Step 1: Extracting double-mutants epistatic divergence

1. For each pair of positions, computed the observed mean relative activity (RA) across all double mutants
2. Store the observed RA in a matrix
3. Plot this matrix as a heatmap
4. Rank these double mutant interactions, and look at the top 5

Step 2: Extracting double-mutants epistatic assuming independence

1. Calculate the expected RA of each positional mutant at positions (i,j), calculate the expected activity if mutations acted independently by multiplying the average single mutant RA:
    a. expected_RA(i,j) = mean_RA[i] × mean_RA[j] (multiplicative model)
2. Store this in a matrix
3. Calculate the difference between observed RA (Step 1) and the expected RA
4. Store this in a matrix, plot as a heat map

Step 3: Identify and rank top epistatic interactions

1. Extract the top 10 position pairs with largest absolute epistasis scores (|epistasis| > threshold)
2. For each pair, note:
    a. Sequence distance: |i - j|
    b. Sign of epistasis (positive/negative)
    c. Strength of interaction

Success Check: You have two heat maps, one of the observed average RA for double mutants, and one of the difference between observed and expected. Table of top 10 epistatic pairs with sequence distance

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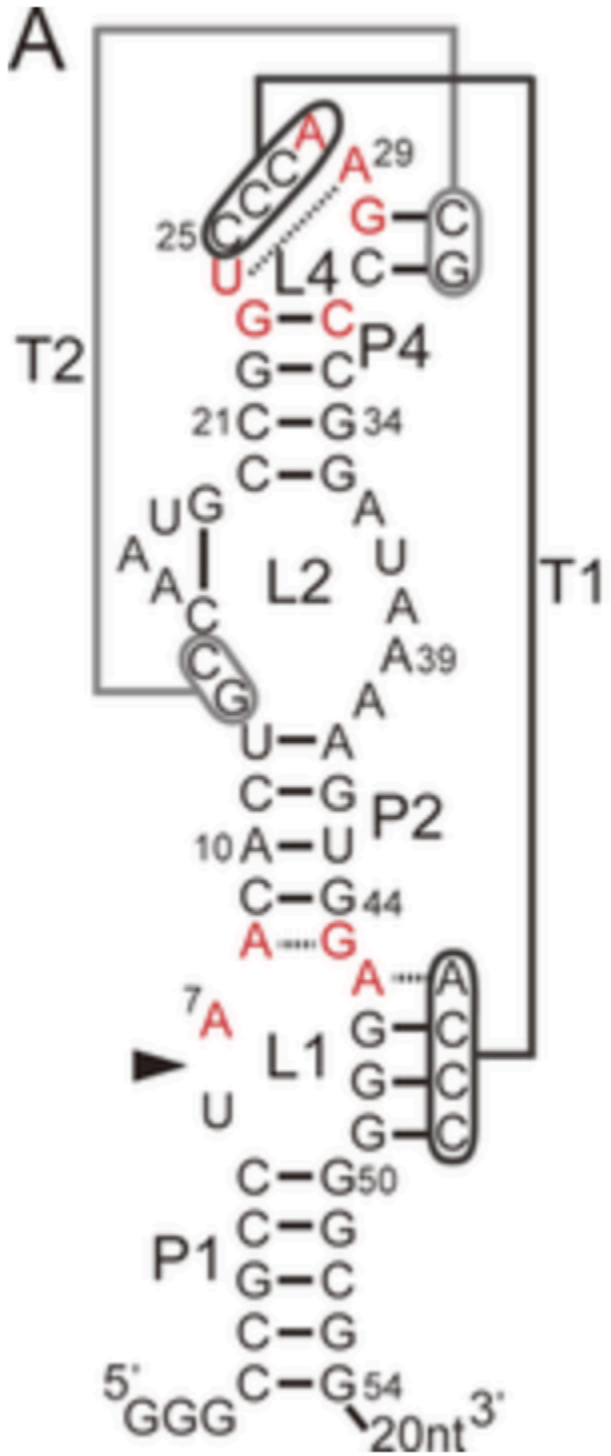
Exercise 4 — Interpretation

Questions: Are there any nucleotide pairs that are epistatically related, but not near each other? Take a look at the secondary structure, are these nucleotides base paired? Are there any nucleotide pairs that are epistatically related but are not related in secondary structure? We used a multiplicative model for our expected epistatic relationships based on single mutant activity, what are the limits of this?
Here is a screenshot of the secondary structure based on a crystal structure, describing the base paired nucleotides. It also includes known tertiary interactions (psuedoknots).

Were you able to recover any of these? Would positive or negative epistatic interactions describe this?



Stretch goal (optional “Dyno”)

Goal: Only identify tertiary, novel contacts

1. Mask ALL known secondary structure base pairs from your epistasis matrix
2. Re-identify top 10 remaining epistatic pairs → pure tertiary contacts

Question: Do these align with known tertiary interactions? (look up pseudo-knot!)

Deliverables

Please submit:

1. Completed notebook (.ipynb)
    - File name: lastname_firstname_RID_xx_code.ipynb
    - Must run top-to-bottom without errors.
    - Includes:
        - Data loading and basic inspection.
        - Single-mutant sensitivity calculations and plot.
        - Double-mutant epistasis calculations, heatmap.
        - A short interpretation section (Markdown).
        - Logbook entry